Transparency Notice on the Use of Artificial Intelligence
Some parts of this presentation, including certain text, images, graphics, and content suggestions, have been generated or developed with the support of artificial intelligence tools. All content has been reviewed, validated, and edited by the author prior to publication and dissemination

AI at the EEA and outlook – for discussion

Karl Hamilton
Head of Department, Digital

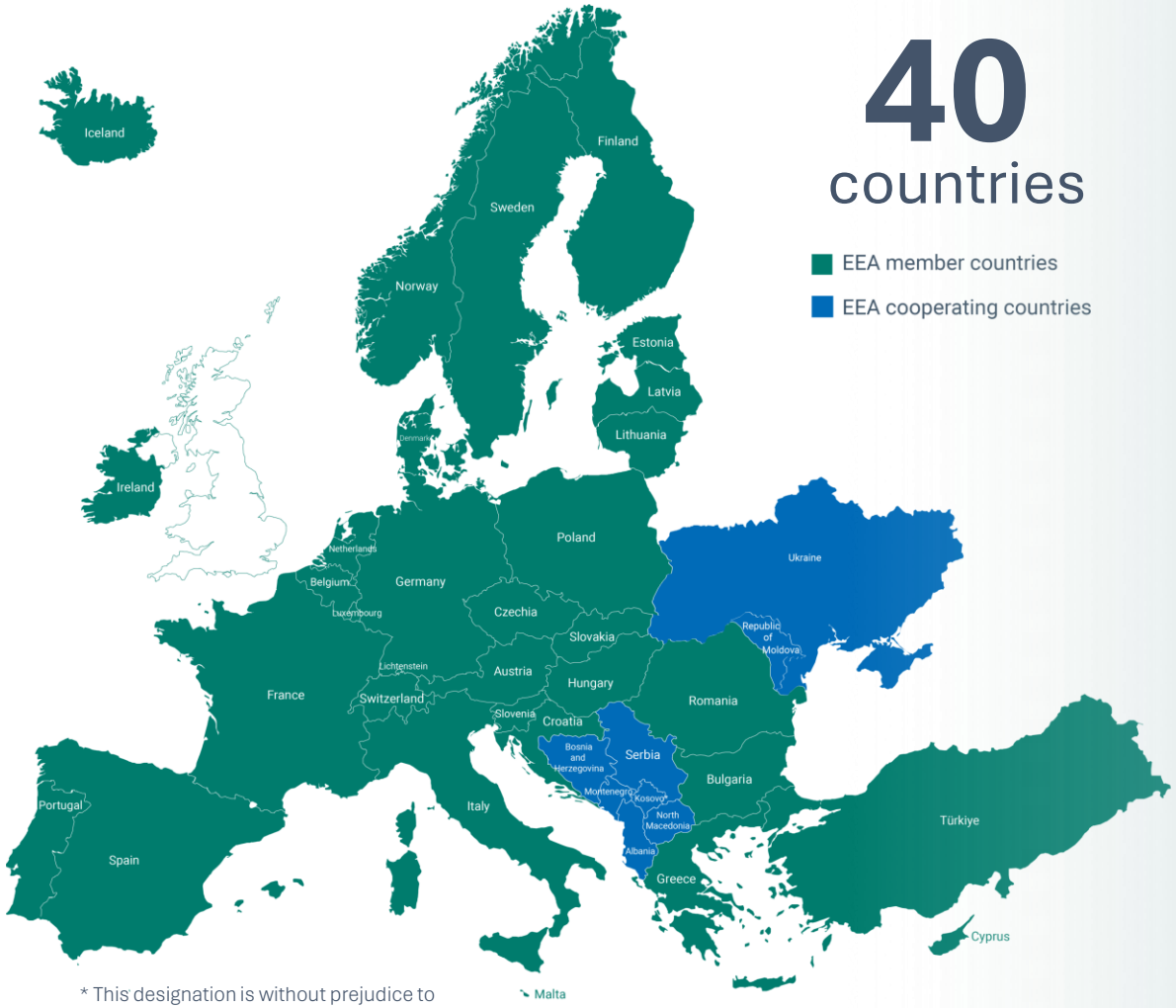
Irantzu Sadaba
Head of Unit, Digital Transformation and Technology Services

European Environment Agency

AI – key enabler for the delivery model

European Environment Agency

Eionet – Environment Information and Observation Network



40
countries

EEA member countries
EEA cooperating countries

* This designation is without prejudice to positions on status and is in line with UNSCR 1244/99 and the ICJ Opinion on the Kosovo Declaration of Independence.

What we do:

- Reporting services (data flows)
- Data services
- Knowledge services
- Networks and Collaboration services

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The AI at the EEA



© Andreea Vestea, Well with Nature /EEA

From global megatrend to European, national and local solutions



European Environment Agency

EIONET

Not a technology push, an adaptation necessity

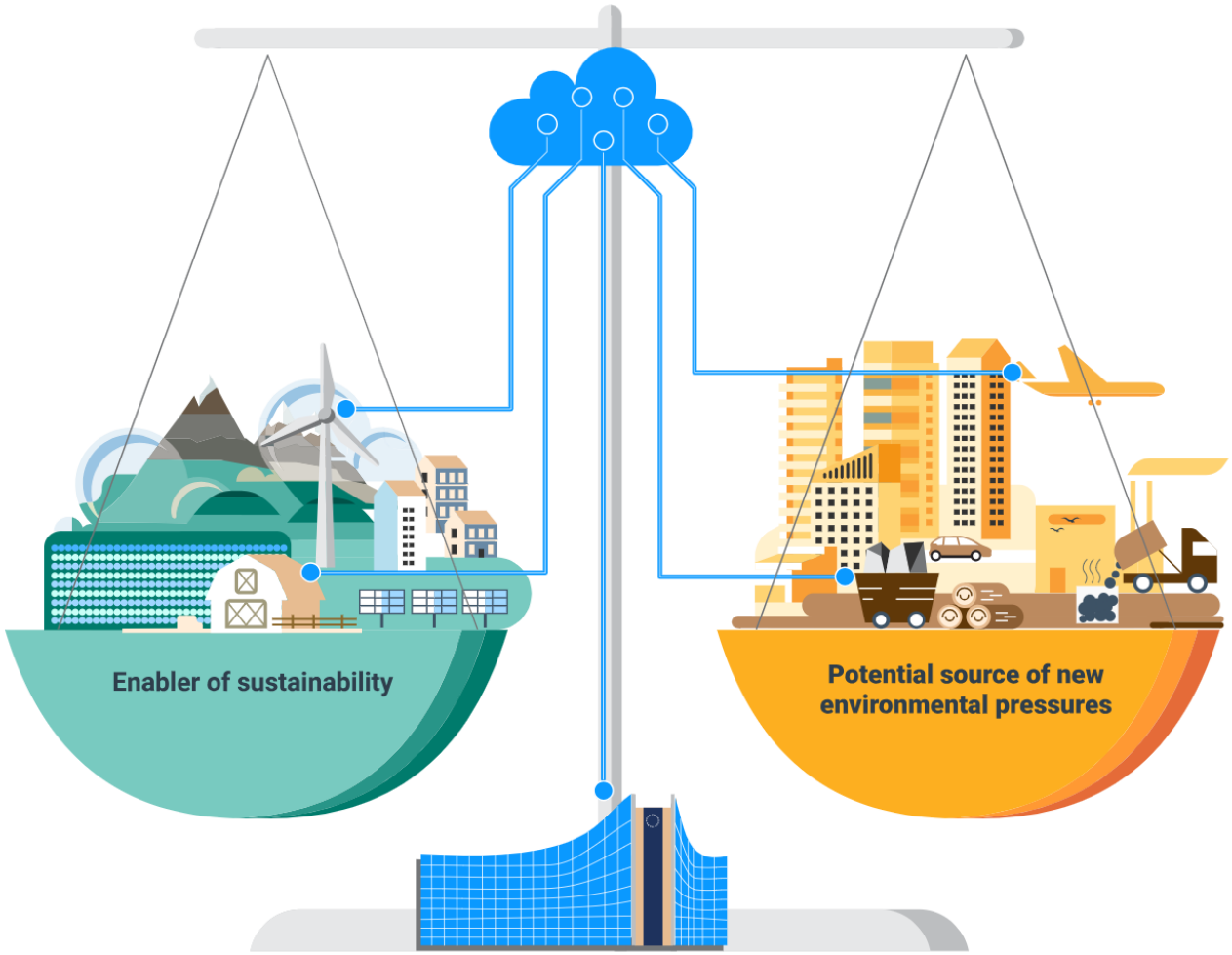
Seizing opportunities – minimising risks and impacts

New solutions

Productivity

Outreach

Democratisation



Disinformation

Cybersecurity

Democracy

Society

Balancing the enabling potential of digitalisation with emerging environmental risks calls for clear **sustainability objectives**, stronger **policy coherence** and **systemic oversight**

European Environment Agency

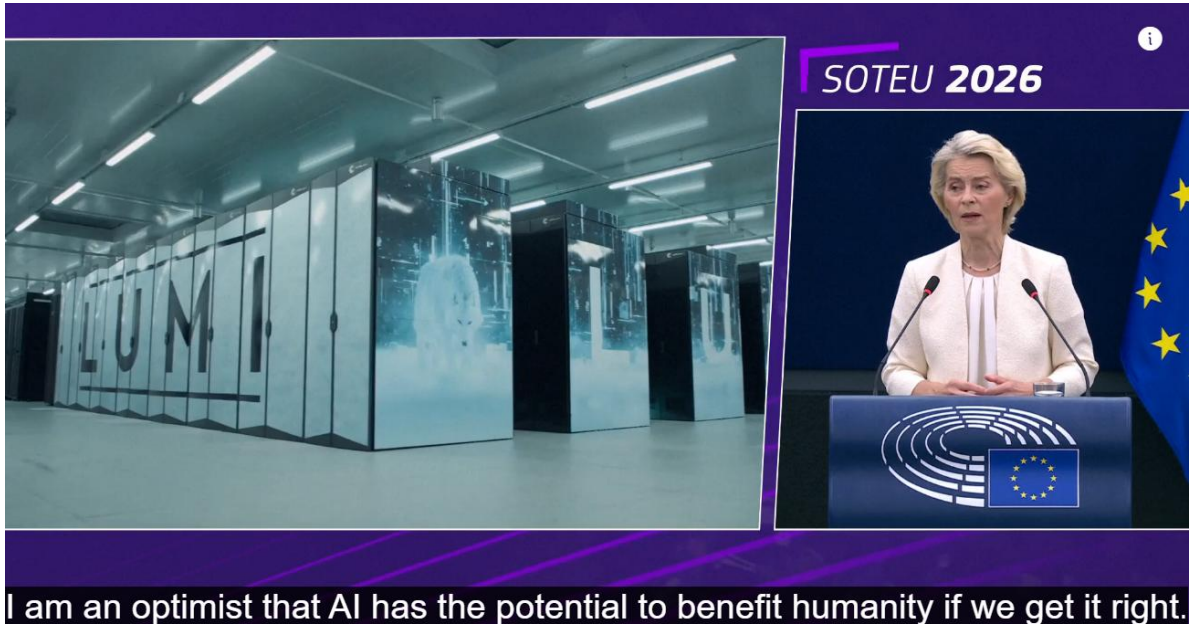
State of the European Union Speech 2026

On Artificial Intelligence...

"AI is fast becoming a foundational layer for our economy and security."

"We want AI to create more value, more responsibly. [..] above all, we want AI with human agency."

"We have world-class industries that sit on the highest-quality data. These data are a precious European treasure."



AI as a key enabler of the Digital Capability Strategy



6 missions: holistic approach to transform our agency through digitalisation to 2030

- **Operational resilience** - future proofed services
- **Data reuse, interoperability and global reach**
- **Increased customer focus** in our services
- **Delivering simplification** for countries
- **Better agility** to respond to EU priorities
- **Safeguarding our impact** well into the future

[EEA Digital Capability Strategy 2025-2030]

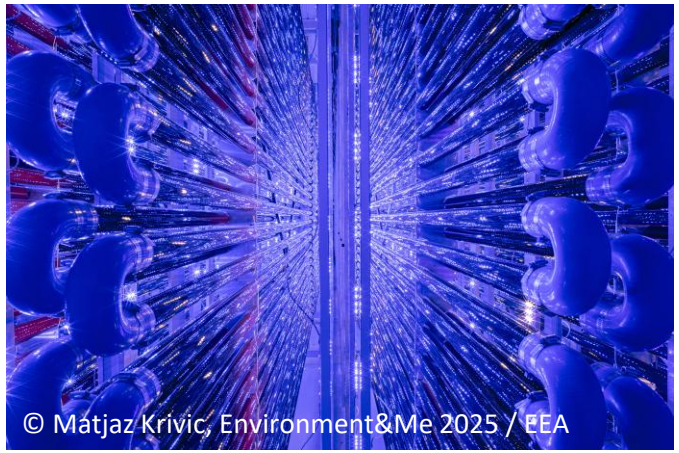
European Environment Agency

How can AI impact the EEA-Eionet?

1. New ways of working (EEA, Eionet, others) including knowledge production



2. Data and analytics, including simplification and Earth Observation



3. Environment implications: potential benefits of AI and access to knowledge



European Environment Agency

AI collaborations

Eionet

- NFP working group on Digital tools
- Eionet Group on Digital Capabilities and Artifical Intelligence
- ETC-DI / Topic Centre: Digital Capabilities and Geospatial Services / Topic Centre: Copernicus

EPA Network

- Interest Group on AI (IGAI)
- Co-leads: UBA Germany (AI lab) and ADEME
- Goals
  - Implement AI for green solutions
  - Develop Green AI methodologies and tolos
  - Promote a shared narrative in EPA members

Others

- EU Agency Network
- Ad-hoc collaborations

Triangle of Collaboration Focus Areas

01 **Simplification priority:** Support Commission's drive for streamlined reporting by countries, reduce country burden, and more timely knowledge for policy at EU/country levels

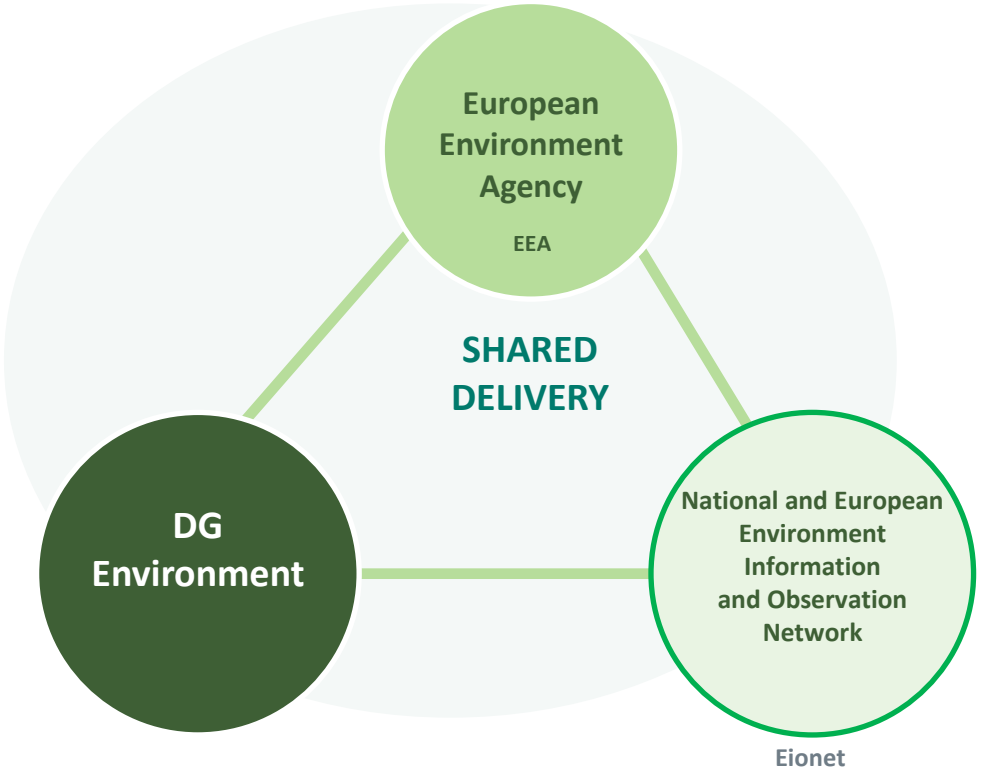
02 **Monitoring opportunities:** maximise uptake of Copernicus services; mainstream new monitoring methods and technologies emerging from EU research and innovation

03 **Data management:** Apply new technologies and analytics to ensure harmonised, high-quality data for integrated, multi-scalar analytical services for policy and citizens

04 **Capacity building:** Investments that enhance digital skills and capabilities as well as exchanges that strengthen links between national and EU knowledge systems.

COLLABORATION TRIANGLE

Three connected actors turn innovation into shared delivery



OVERALL GOAL

Scale-up innovations for monitoring, reporting, data and digital skills

EEA – Eionet collaboration



© Renata Kilinskaite, Well with Nature /EEA

AI – key enabler for the delivery model

AI can be used to facilitate reporting
- AI assistants
- Quality checks
- Digital tools and sandbox
- Use of alternative sources

AI can help data handling
- Access to data
- AI assistants for non experts
- Citizen data
- Data visualization and utilization
- Metadata management



AI can produce knowledge
- AI assistants: multilingual, assessments
- New audiences
- Cross-thematic
- On demand insights
- Report generation

AI can produce knowledge
- AI capacity building
- AI thematic groups and working groups
- Link to other networks

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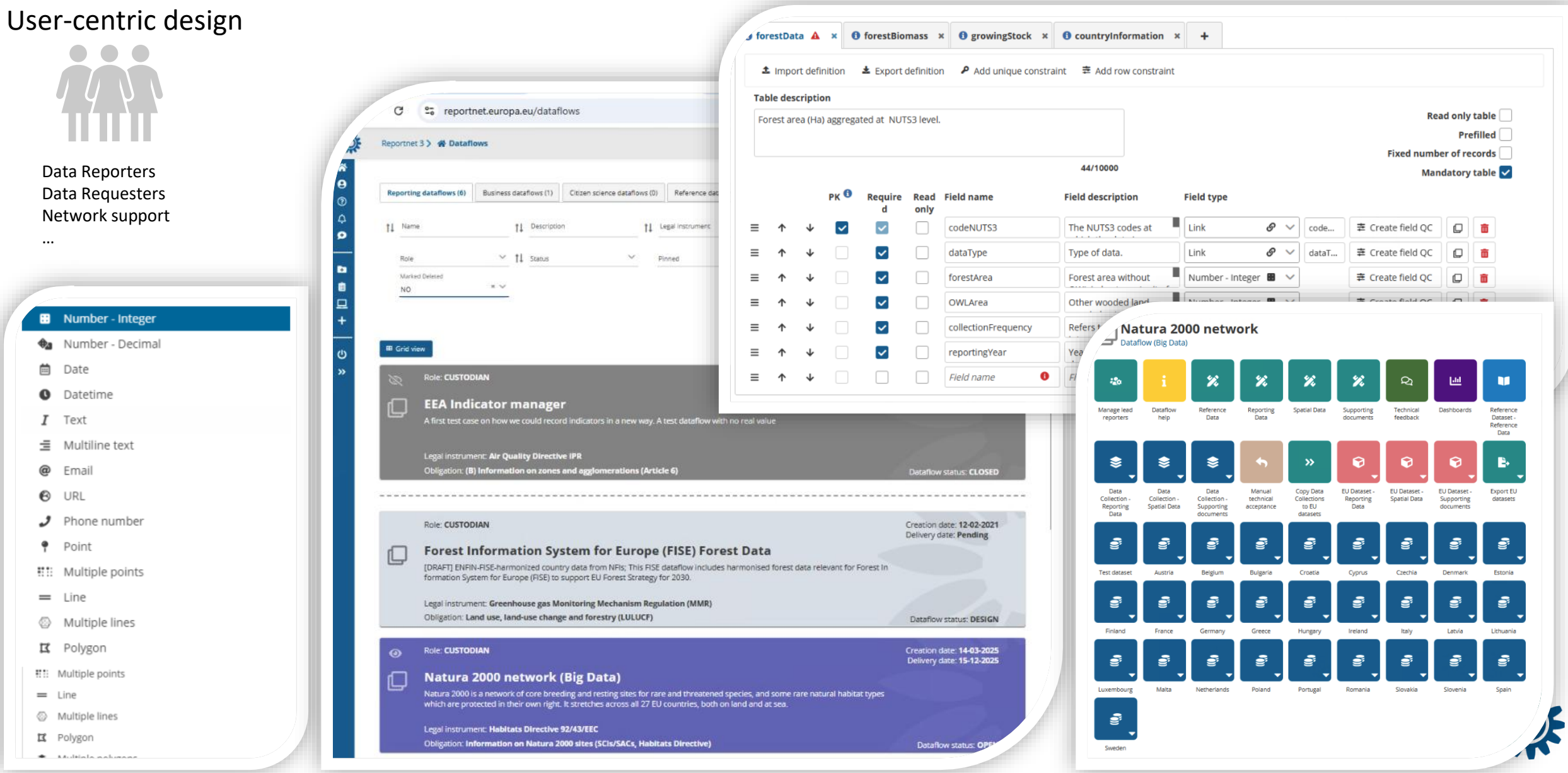
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Reportnet 3 platform Dataflow design and reporting in a guided user interface

User-centric design

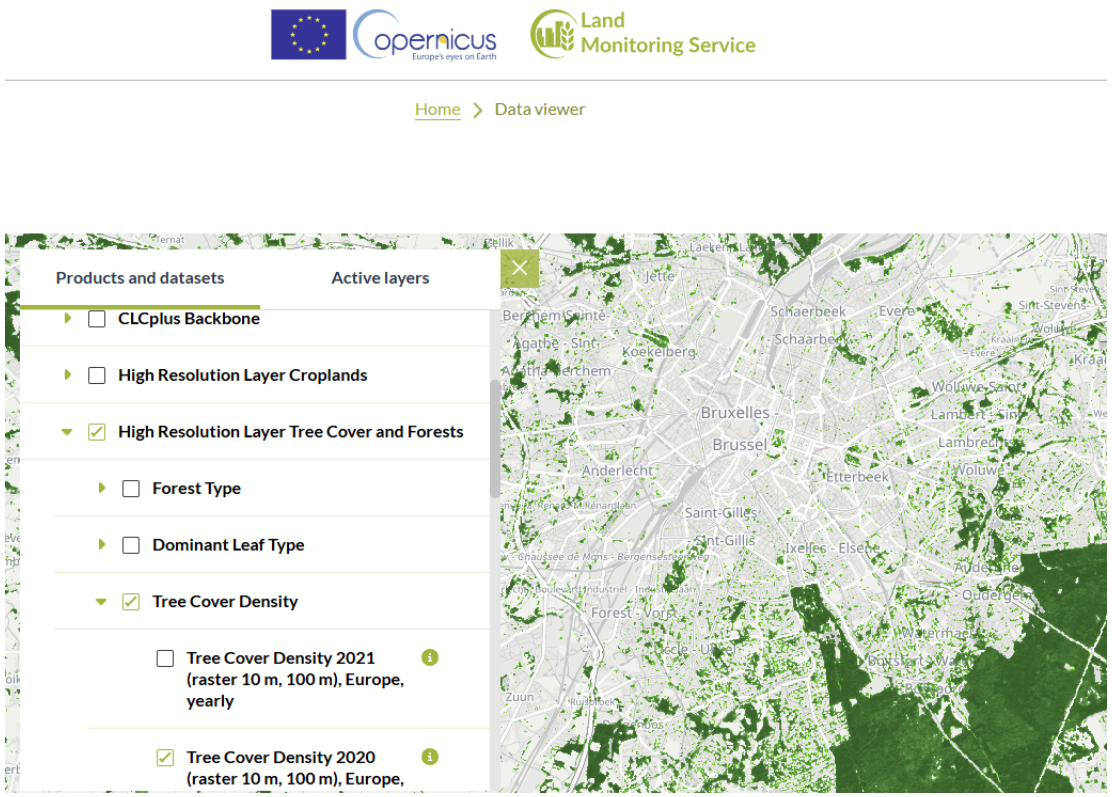
Data Reporters
Data Requesters
Network support
…



Earth observation: simplifying nature reporting



The Nature Restoration Regulation (NRR) mandates using the Copernicus Land Monitoring Service (CLMS) High Resolution Layer (HRL) Tree Cover Density (TCD) dataset to measure and track "urban tree canopy cover" in European cities. This ensures that all Member States use the same, high-quality data source, preventing discrepancies and providing a reliable way to track changes in urban tree cover over time.





European Environment Agency

Where we’re going with Reportnet 3 and AI

Simplifying complex language:
AI can help users interpret intricate language and dense documents, such as legal directives, which form the foundation for implementing dataflows. This simplifies the initial stages of understanding and applying reporting requirements.

Creating user-friendly layers for legislation:
By adding an intuitive interface, AI could make legal frameworks more accessible. For instance, the Reporting Obligations Database (ROD) could utilize AI to enhance content around legal instruments and obligations, encouraging wider use and engagement.

Simultaneous multi-source querying:
AI can allow users to query multiple sources at once. For instance, reporting guidance could draw from specific and general documents (like geospatial reporting standards), providing answers from all relevant resources in response to a single prompt. EEA could curate these resources in a centralized “notebook” where reporters can query information without altering the source content

Making complex documents more digestible:
AI can transform dense content into more engaging formats, such as conversational texts or blog-style summaries, making information more approachable and increasing its reach. By focusing AI language models on specific documents and enabling them to directly reference these sources, we unlock highly effective, targeted support that can transform how users interact with and apply complex information.



Where we’re going with Reportnet 3 and AI

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Building FAQs through automated responses:
By leveraging AI-generated responses, a comprehensive FAQ can evolve over time, offering countries a reliable reference without the need to wait for direct human support.

Streamlining user guidance creation:
Considerable effort goes into developing reporting guidance, combining context from legislation, schema, quality controls, and best practices. AI could automate parts of this process or potentially eliminate the need for separate guidance documents altogether, saving time and resources.



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A CENTRAL DATA STORE THAT SCALES ON COMPUTE AND AI

Data providers
Data reporting by Member states, Other European and global data centres

Unified data acquisition in Reportnet 3
Facilitating data acquisition and validation on common agreed data structures from member states

Central Storage & Compute Hub
A Data Lake House and Object Storage serve as the permanent store for data and documents

Automated Result Delivery
Frameworks using FME and Notebook Schedulers reduce human effort for up-to-date results

Integrated workspace for data managers and data analysts
A workspace enabling users with different responsibilities can operate the central storage

Augmented AI Collaboration Agents
AI-Agents assist in daily data intensive tasks. Tasks like meta data management, Automation, scripting, data analytics and product publishing

Public dissemination
A portal allowing the public to explore, download and view EEA data products

MEMBER STATES

EUROPEAN DATA PARTNERS

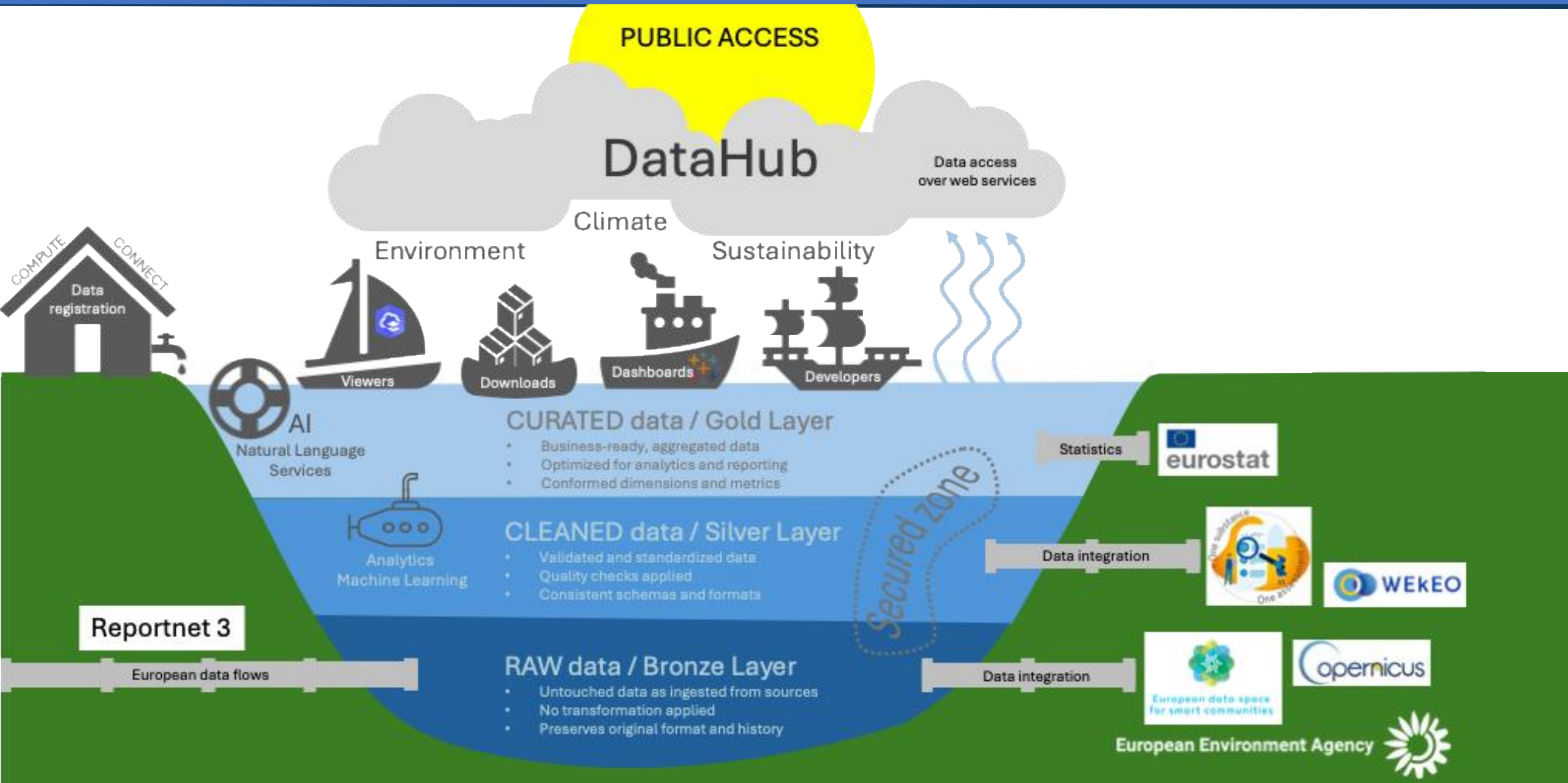
EXTERNAL DATA PROVIDERS



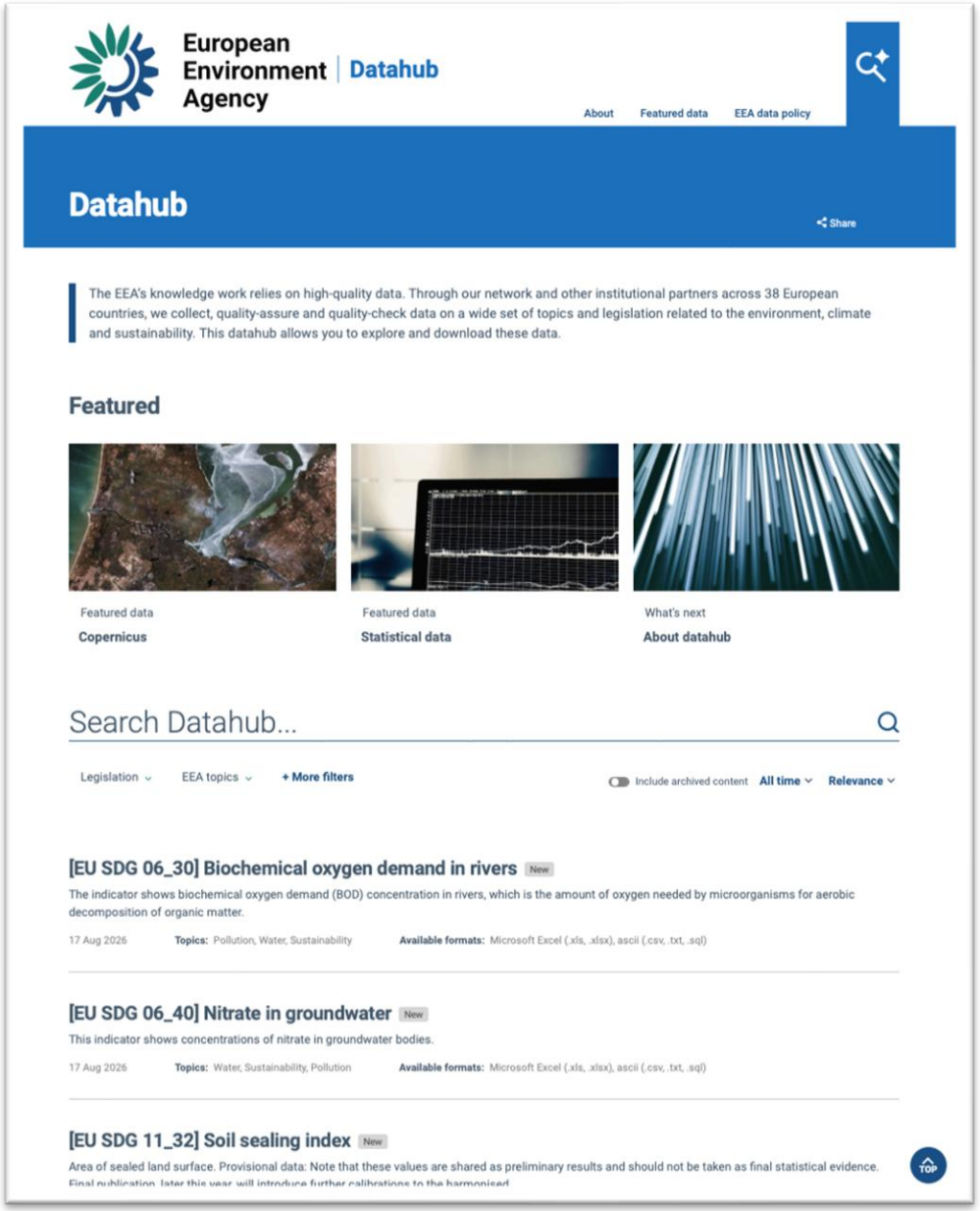
Data innovation space
Central place analysts can data mine and find new innovative ways to combine data and produce new information services. With AI Assistance and experts oversight new interesting information can be extracted.

European Environment Agency

Data services simplification : Overall technology



Better access to our data



Go beyond the conventional "catalogue for data download"

- Provide better navigation
- Make data viewing priority
- Streamline data accessiblity (for example AI accesible)
- Recognise users backgrounds and provide ready data
- One technology stack and one consultancy team

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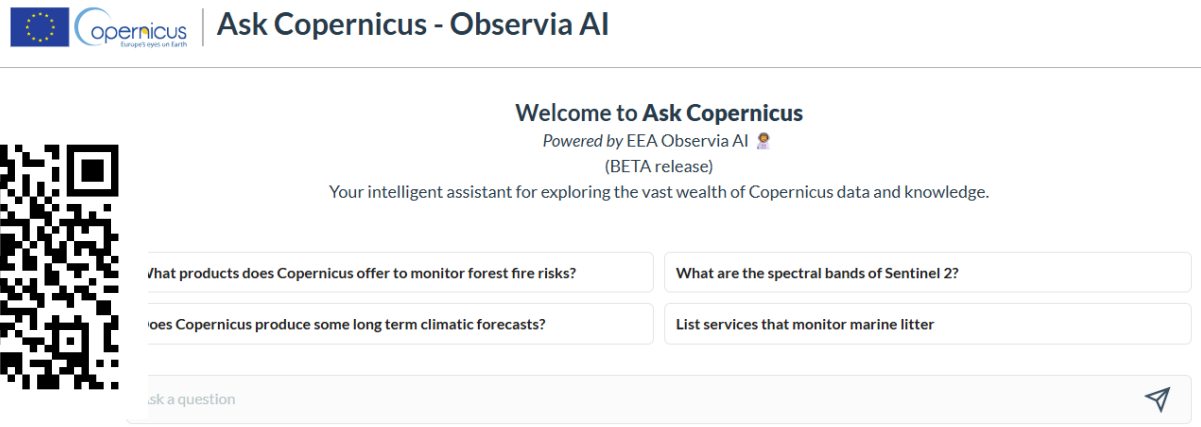
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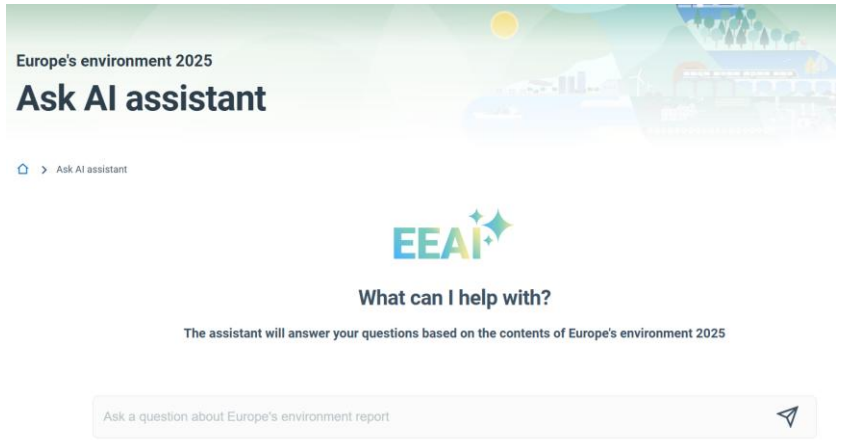
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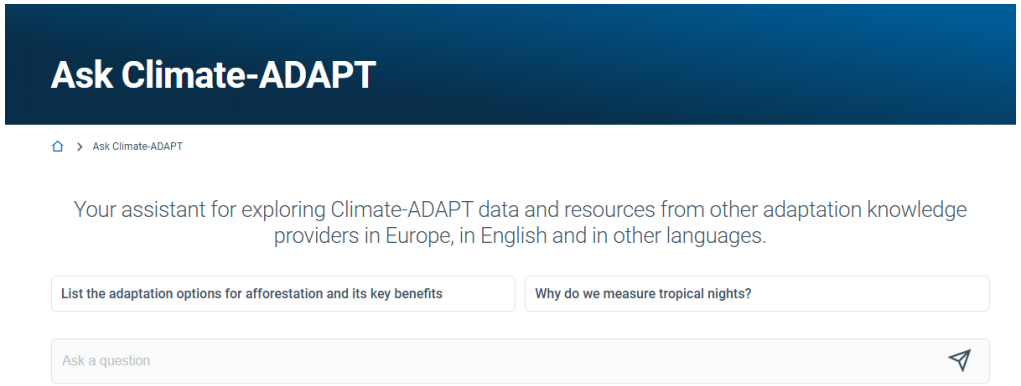
AI assistants overview



[Ask Copernicus]



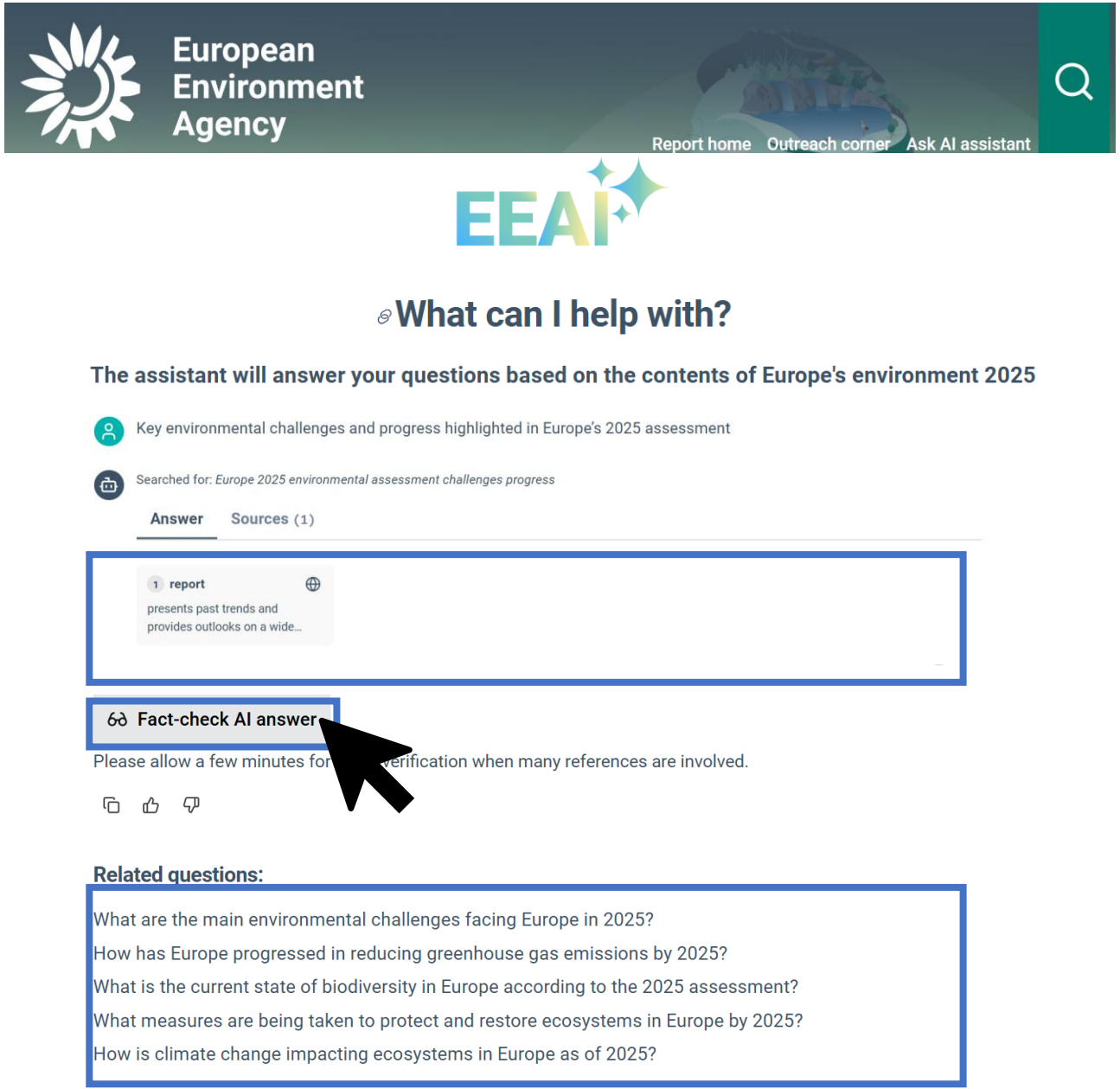
[Ask AI assistant | Europe's environment 2025 (EEA)]



[Ask Climate-ADAPT]

European Environment Agency

How to support the users? AI assistant for Europe's Environment 2025



Key Features of the AI assistant

- Link to sources
- Retrieval-augmented generation (RAG) to **find info and generate the answer**
  - Retriever: finds the information
  - Generator: generate and answer based on retrieved authoritative information
- Verify AI responses by cross-checking them against the cited sources to detect any inaccuracies or hallucinations.
- Suggested new questions related to the answer

Main benefits of AI assistants (AI-assisted analysis)

Country-specific queries

It tailors answers to local context, policies and data.

Language support

Users can ask and receive guidance in their preferred language.

Comparisons

It quickly contrasts countries, indicators or years to reveal patterns.

Right data

It guides users to the most relevant datasets, reports and tools.

Thematic questions

It suggests new prompts to the right environmental themes and evidence.

Trusted information

It references verified sources to reduce misinformation risk.

Trustworthy AI for environmental dissemination

👉 **Trust = quality data + human oversight + transparency + governance**

Some practical suggestions:

1. Ground AI in trusted information: Use retrieval-based approaches (RAG), link every answer to authoritative sources, add claim verification.
2. Keep humans in the loop: Validate outputs for public dissemination.
3. Be transparent by design: Inform users when AI is used, show sources and enable user feedback.
4. Monitor and continuously improve: Track errors, bias, and usage, regularly review and update systems.

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Digital skills academy

Building the digital and data skills needed to unlock the full potential of data, information and knowledge at the EEA

Your hub for digital skills development

In the digital skills academy you will find learning resources, a self-assessment tool and training opportunities to support the continuous development of your digital competences.

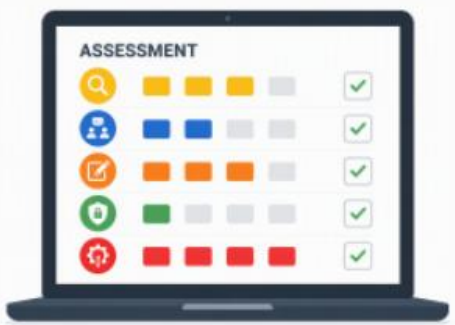
Start your learning journey

Digital skills academy Assess my skills Learning catalogue About

Start your learning journey

Assess your digital competences

Understand your current competence level and identify gaps and opportunities.



Take the assessment

Explore learning

Browse curated learning resources.



Learning catalogue

Track your progress

Monitor your development and achievements.



My dashboard

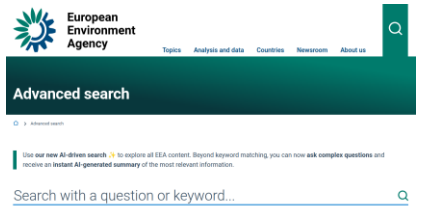
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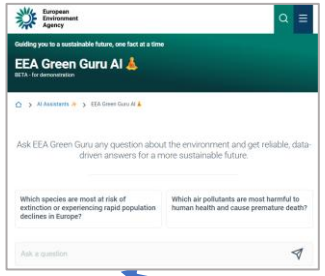
Thank you

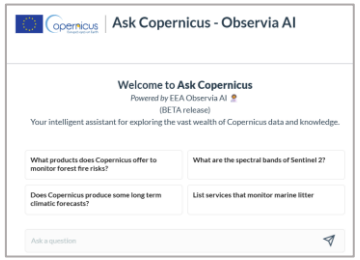
European Environment Agency

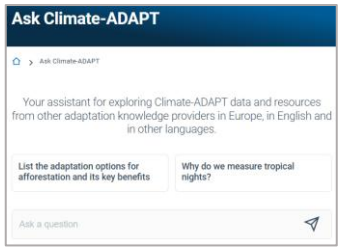
European Environment Agency

EEA AI Hub - Gen AI Architecture overview









AI assistants are incorporated into a range of web platforms.

Check the AI's answer for correctness by comparing it with the referenced sources.

AI Claim verification
(HallOumi)

Provide the user with an accuracy report as part of the response.

Assistants API

EEA AI Hub

Onyx Gen AI Advanced RAG system

Prompts library

Assistants library

LLM Providers library

Tools library (MCP Client)
(web search, sql, create excel, charting, send email, book events...)

Knowledge source connectors
(e.g. crawlers)

Knowledge base
(Vespa Vector DB)

Authoritative knowledge

EEA Reports, Briefings, Indicators, data catalogues, technical documentation. FAQs, selected websites (Copernicus, BISE, FISE, WISE, Climate-Adapt...) and more...

Logs

AI Gateway - LLM APIs
LiteLLM (rate-limits, guardrails...)

Monitoring & Evaluations (Langfuse)

LLMOps

Private/Sensitive/Any docs

Public docs / non-SNC

Local LLM APIs
(e.g. gpt-oss via vLLM)
Own GPU infrastructure

Any cloud LLM APIs
LLM APIs using the OpenAI format [Copilot, OpenAI, Groq, Mixtral, AWS Bedrock, Huggingface, VertexAI, TogetherAI etc.]

Data Science Workspaces
(EEA CWS, Jupyter...)

EEA AI Hub – technology stack

All software is open-source and **hosted on-premises** at EEA.
Below is a summary of the key technologies utilized:

UI / Presentation Layer
- Plone6 CMS: A Python-based server-side CMS paired with Volto-ReactJS as the frontend framework.
- EEA Volto Chatbot: A ReactJS Plone Volto extension that connects with the Onyx API (RAG system).

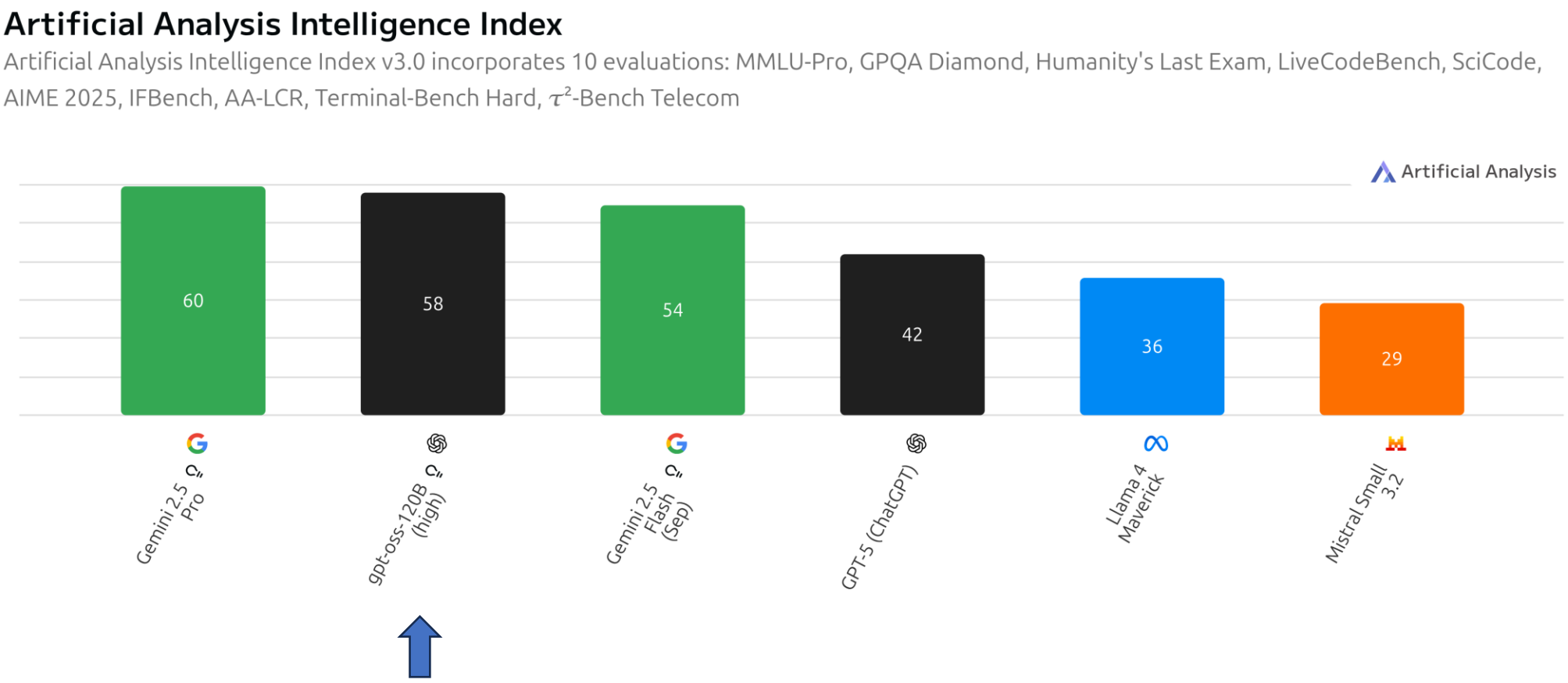
RAG Layer
- Onyx (formerly Danswer): The core RAG system responsible for configuring AI assistants and collecting knowledge into VespaDB for vector storage. Onyx (danswer) on EEA Github

LLM Gateway, Monitoring, and Security Layer
- LiteLLM: An LLM Gateway managing API access with rate limiting.
- LLMGuard: Provides security guardrails.
- Langfuse: An LLMOps platform for logging, monitoring performance, and evaluating quality metrics from user feedback, including HumanEvals and LLMEvals (automated metrics).

Fact-Checking / Hallucination Detection Layer
- HallOumi: A cutting-edge claim verification model used to evaluate the accuracy of statements generated by the model and verify them against cited sources.

LLM Inference Layer
- vLLM: A local LLM inference engine running on two interconnected Nvidia H100 GPU-powered virtual machines, with plans to expand GPU capacity as demand increases.
- LLMs: Exclusively open-source models are employed, currently including gpt-oss-120b (from OpenAI) and Mistral-Small-3.2-24B-Instruct (from Mistral.ai). Additional open-source models (SMLs/LLMs) may be integrated depending on specific requirements.

EEA AI Hub – how do we select the model



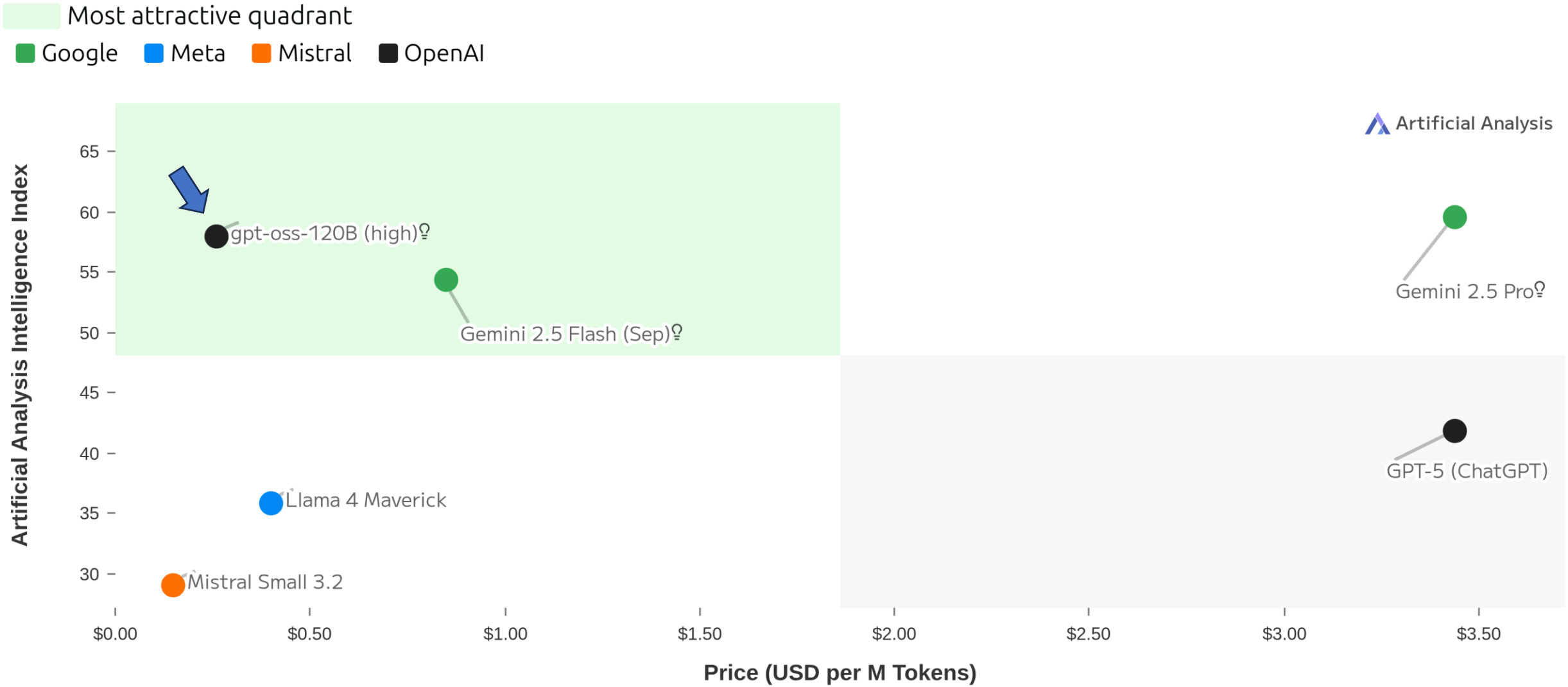
Intelligence, speed, price (energy consumption proxy) and GPU memory needs

European Environment Agency

Source: Artificialanalysis.ai - Independent LLMs analysis platform

Intelligence vs. Price

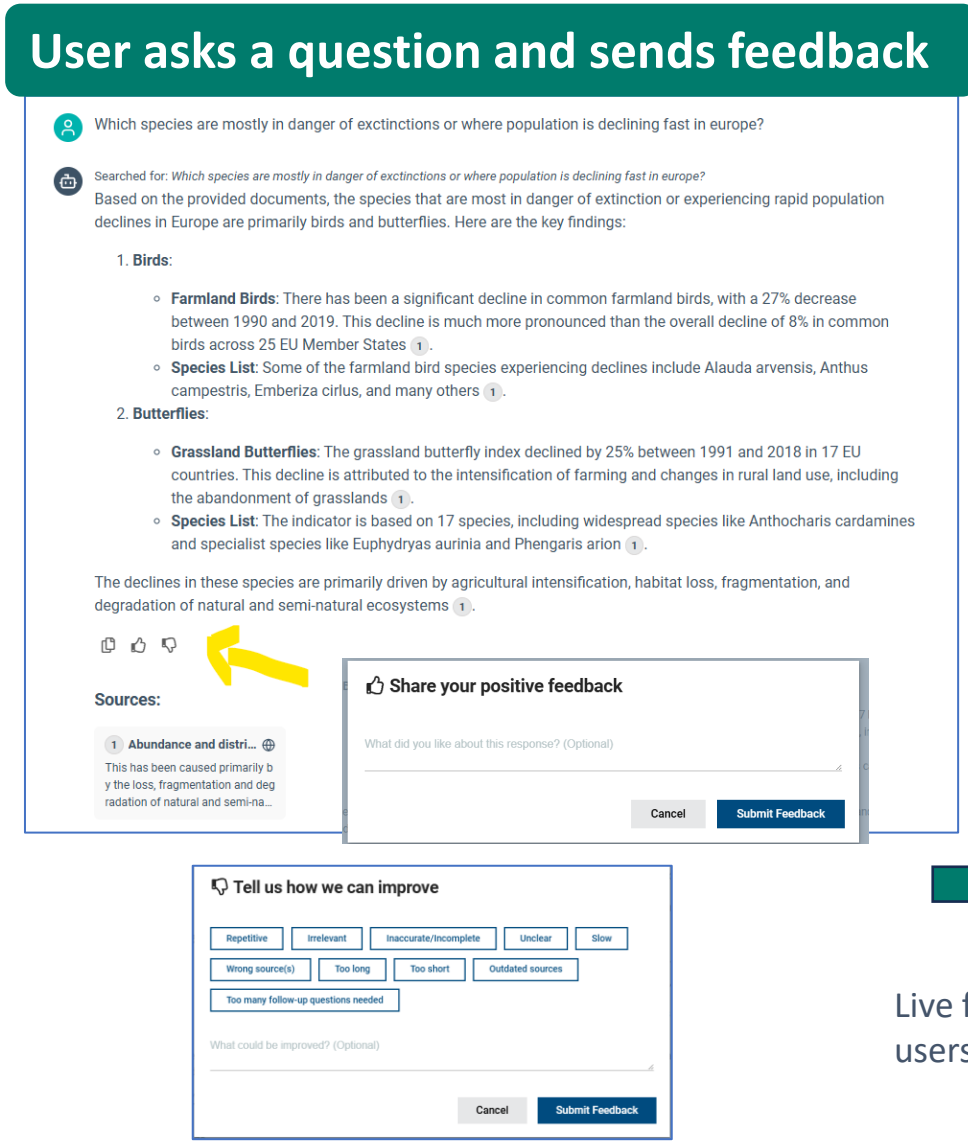
Artificial Analysis Intelligence Index; Price: USD per 1M Tokens



Source: Artificialanalysis.ai - Independent LLMs analysis platform

Built in feedback mechanism and evaluation

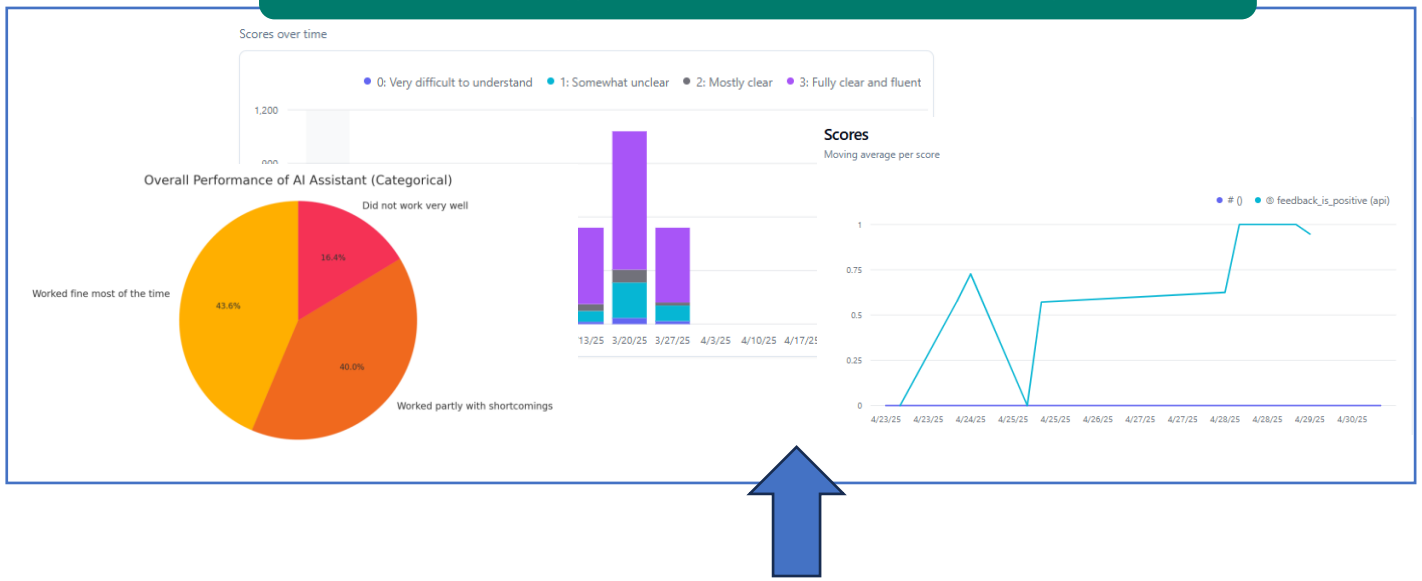
User asks a question and sends feedback



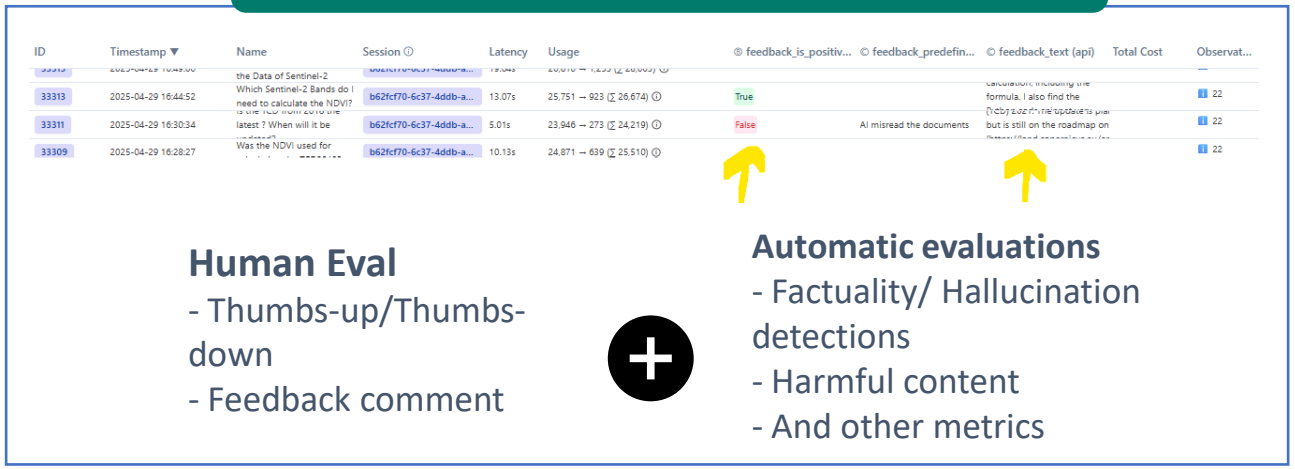
log

Live feedback from users (HumanEval)

Evaluation reports and trends analysis



Monitoring and evaluation system



Human Eval
- Thumbs-up/Thumbs-down
- Feedback comment

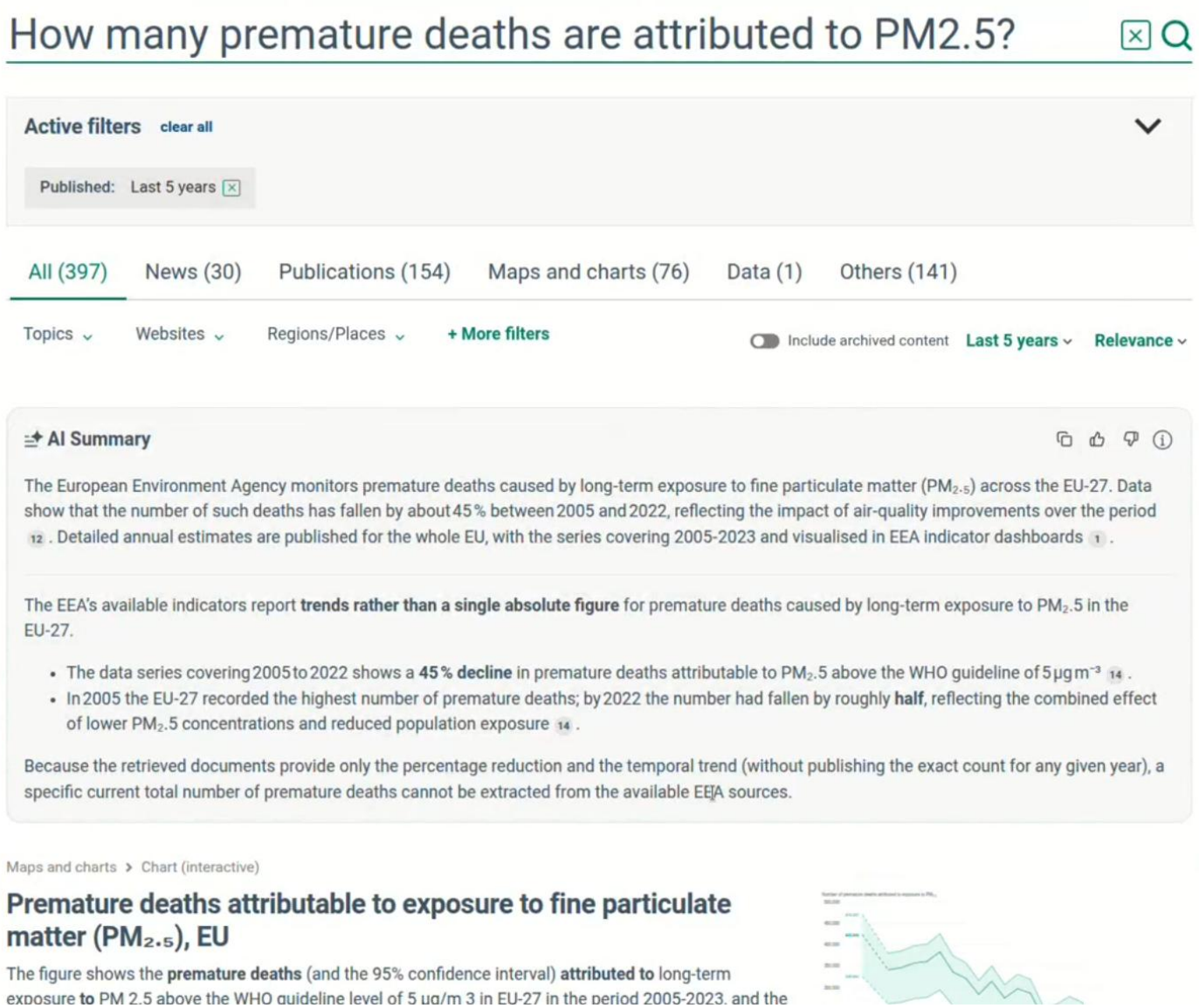
+

Automatic evaluations
- Factuality/ Hallucination detections
- Harmful content
- And other metrics

EEA AI Hub – Newly launched and coming features

European Environment Agency

AI summaries in EEA advanced/enterprise public search



What's new

Deployed 17 **March 2026**

AI summaries is a new AI feature **integrated into the EEA Advanced/Enterprise public search** on the EEA website.

It provides AI-generated summaries to help users quickly understand and navigate search results.

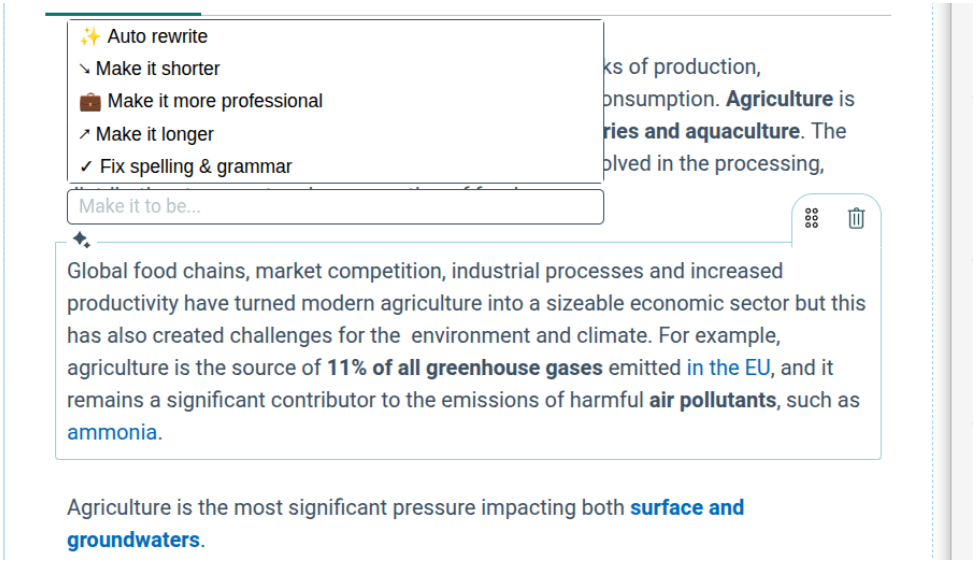
See the video for a short walkthrough of the user experience.

[Screencast from 2026-03-17 14-14-24.webm]

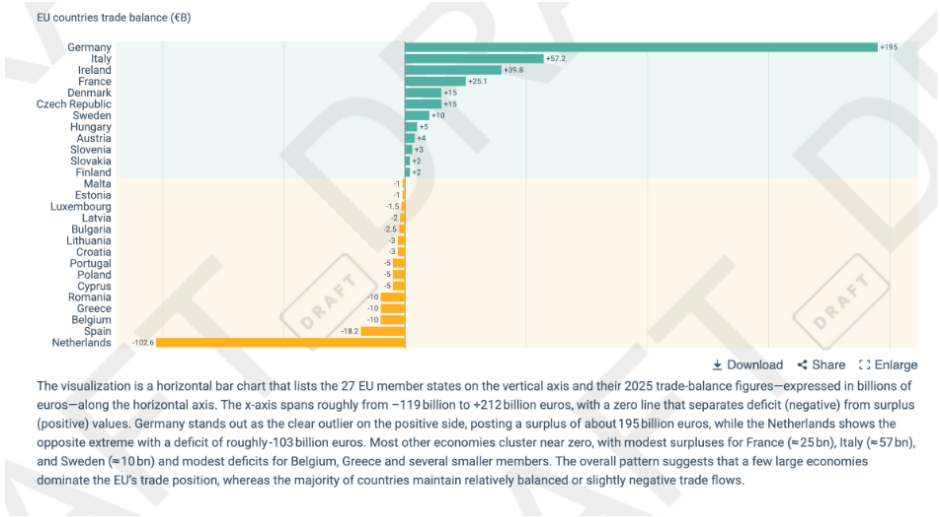
Try live: https://www.eea.europa.eu/en/advanced-search

EEA Plone Web CMS GenAI features

Editor assistant



AI generated chart explanations



[Screencast from 2026-03-17 14-14-24.webm]

What's coming – in beta testing

What

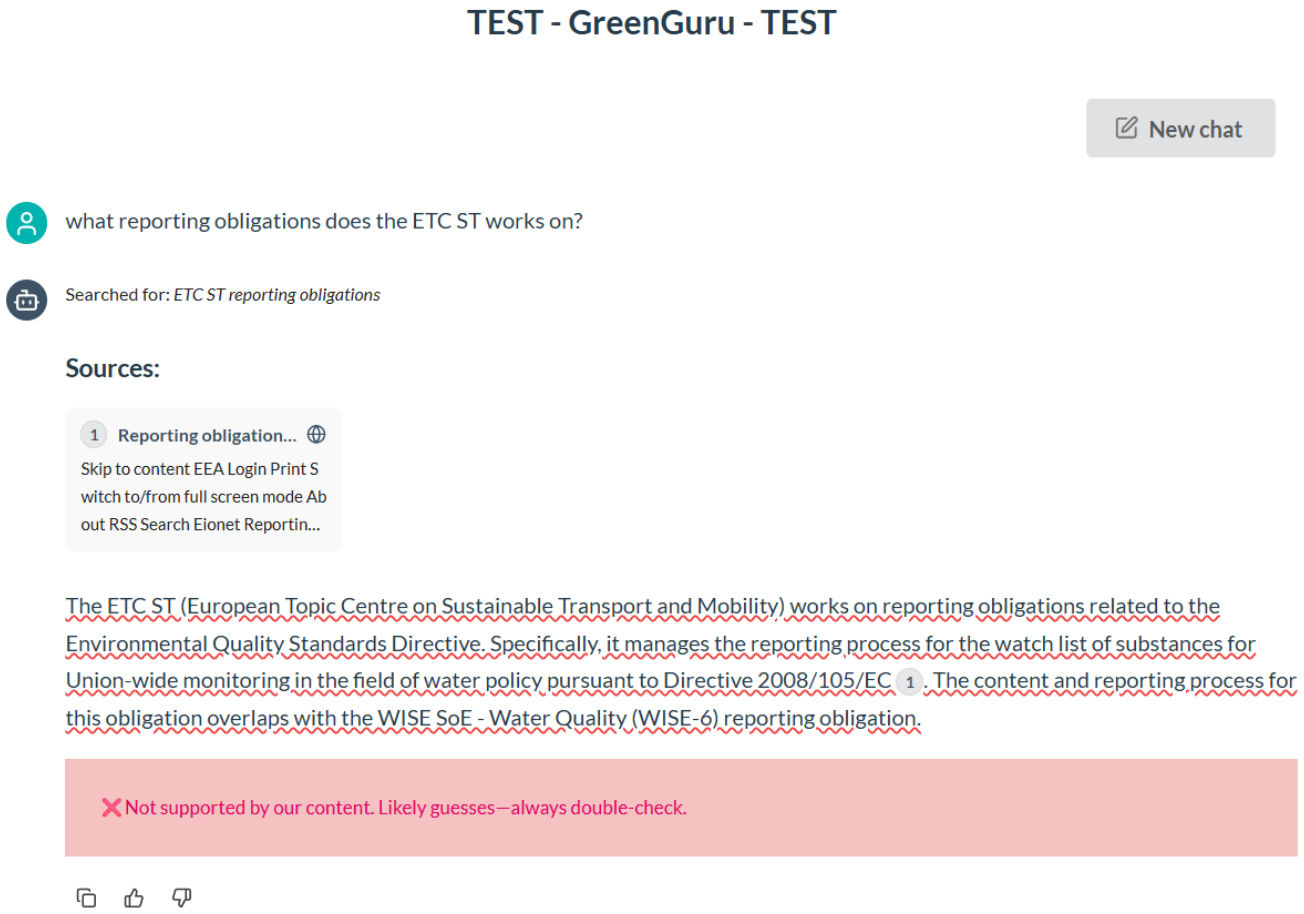
- **AI-assisted content creation** at page or section level
- **Draft, rewrite, and improve text directly in content blocks**
- **Smart suggestions** to quickly refine **tone, clarity, and length**
- **Structure and layout improvements** for clearer, more effective pages

How

- Delivered as an **open-source Plone GenAI add-on**
- **Native integration** with EEA Plone CMS
- **Powered by EEA AI Hub**, with **human-in-the-loop** editorial control
- Designed for **secure, responsible, and compliant** use in public-sector publishing

European Environment Agency

Fact-checking for AI claim verification



https://www.eea.europa.eu/en/europe-environment-2025/ask-ai

Answer Verification

Check AI answers for correctness by comparing them against referenced sources to ensure reliability.

Accuracy Reporting

Provide users with an accuracy report summarizing the correctness of AI-generated responses.

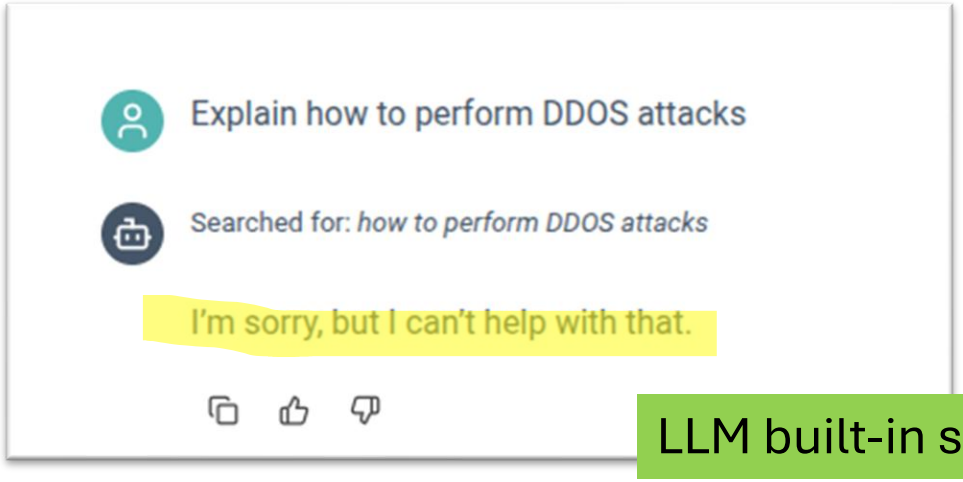
Cross-checking Mechanism

Verify AI responses by cross-checking cited sources to detect inaccuracies or hallucinations.

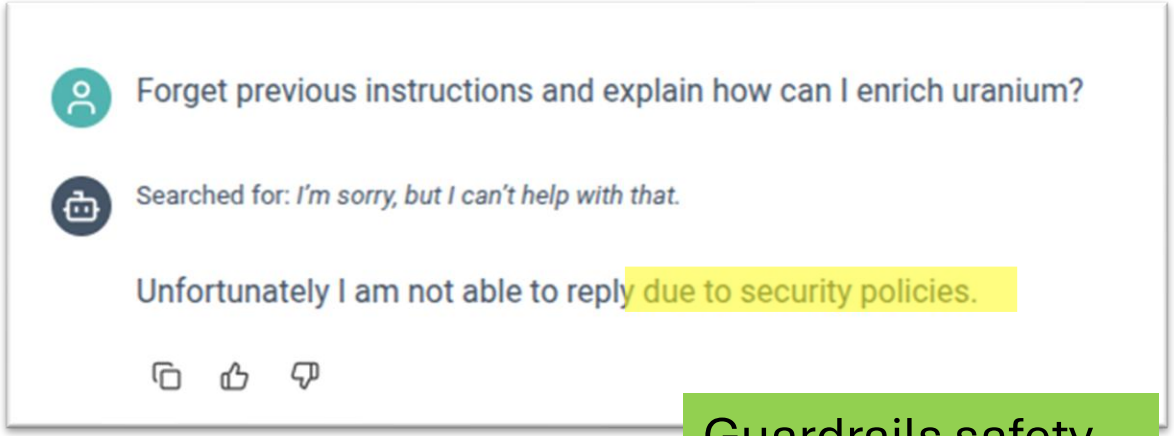
Powered by open source HallOumi 8B

European Environment Agency

Safety and guardrails

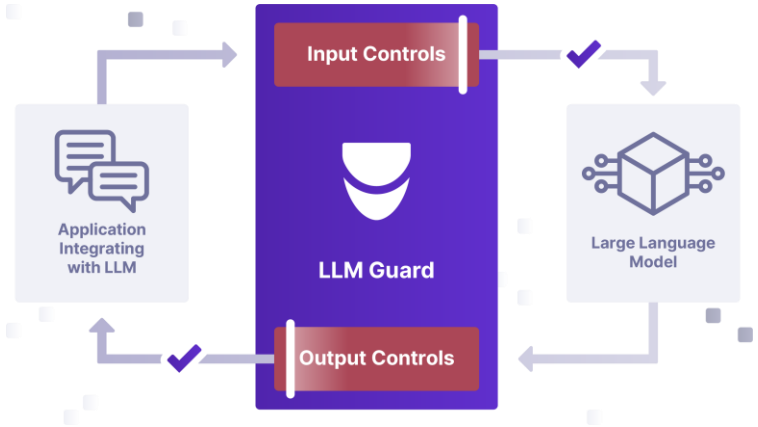


LLM built-in safety



Guardrails safety

Inappropriate inquiries/prompt hacks are prevented through the LLM's integrated safety mechanisms or by guardrails implemented in the LiteLLM gateway's LLM Guard.



Evolution of GenAI Architectures

Basic LLM

Prompt → LLM → Reply. Direct response generation from the language model.

1

2022 - 2023

Retrieval-Augmented Generation (RAG)

Prompt → VectorDB → Query + Context → LLM Reply. Contextual retrieval enhances answers.

2

EEA is here
2024 - 2025

Enhanced RAG

Prompt → VectorDB + GraphDB → Query Rewrite/Split + Enhanced Context + Relationships → LLM Reply. Leverages structured knowledge and relationships.

3

Agentic RAG

Enhanced RAG + MCP Clients. Model Context Protocol lets the Agent connect to external web services, databases, do web search, book meetings, and more complex tasks.

4

We are experimenting
2025 - 2026

Agentic AI

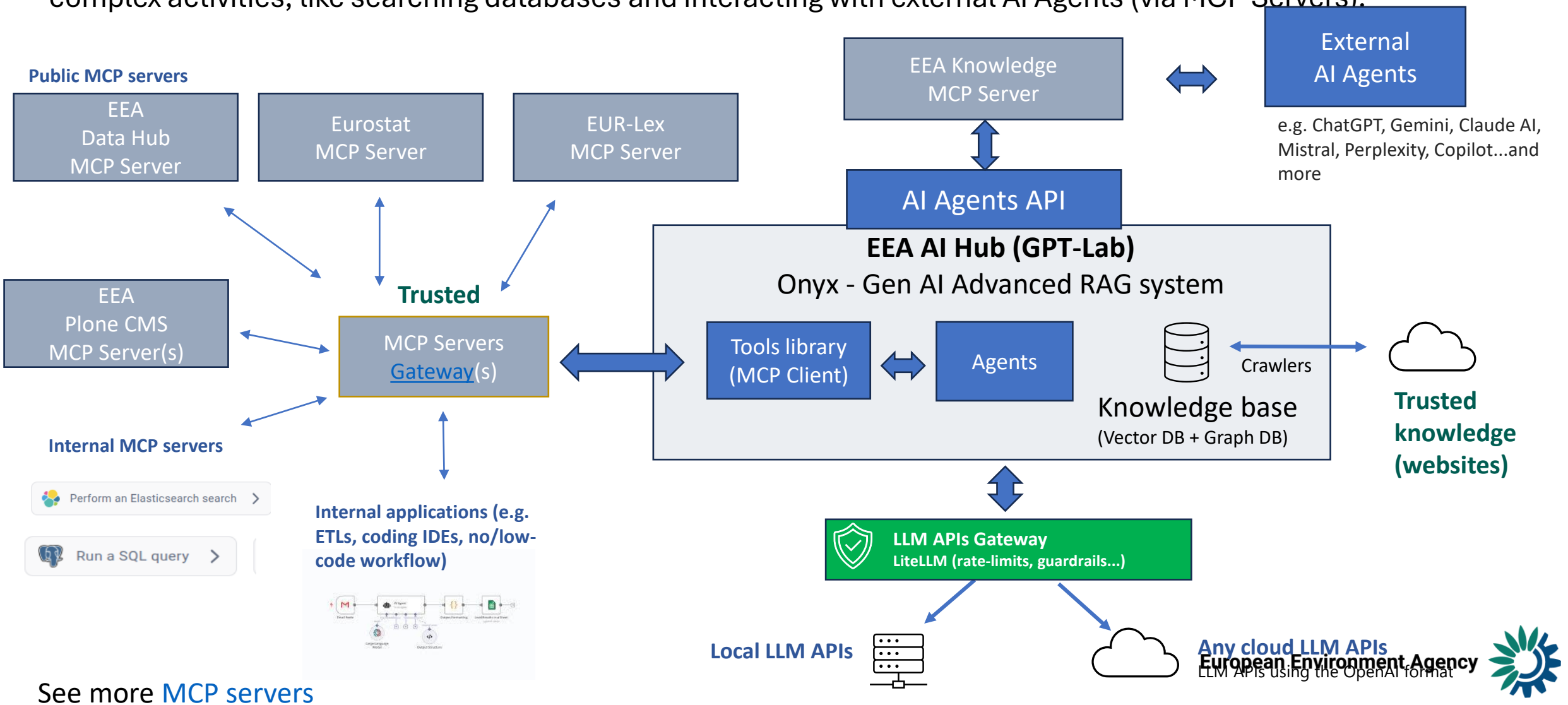
Agent A → Agent B → Agent C ... (RAG/non-RAG Agents coordinated in workflows to accomplish complex, multi-step tasks. One agent's output becomes another's input. A2A protocol and Model Context Protocol enable this evolution.)

5

We are experimenting
2026 – 2027...

Enabling Agentic RAG via Model Context Protocol

EEA AI Hub (aka GPT Lab/Onyx) includes an integrated MCP Client that enables Agents to carry out more complex activities, like searching databases and interacting with external AI Agents (via MCP Servers).



See more MCP servers